

3(a)	any three from: 1 manipulation of, genetic material / gene / DNA or described ; 2 <i>ref. to</i> gene / allele, transfer (into, cell / organism) or described ; 3 expression of gene ; 4 (named) phenotype modified / (named) protein produced ; 5 AVP ; e.g. <i>ref. to</i> gene editing / produce GMO	3
3(b)(i)	any three from: 1 add, healthy / normal / correct / functional, gene / allele, to virus ; 2 inject virus into eye ; 3 gene integrated into (eye cell) genome / DNA ; 4 gene / allele, expressed to produce functioning, protein / enzyme ; 5 AVP ; e.g. adeno-associated virus / AAV voretigene neparvovec (drug)	3
3(b)(ii)	any one from: 1 easily accessible / easy to inject ; 2 low risk of immune response ; 3 (eye small in size so) small amount of gene therapy treatment needed / AW ;	1

Question	Answer	Marks																																							
3(c)(i)	any three from: (overall) improvement in acuity of both eyes or (overall visual) acuity of both eyes becomes more negative ; - treated eye, has more improved acuity / has more negative (visual) acuity ; ora <i>ref. to</i> anomalous result at month 9 / result that does not fit the pattern at month 9 ; - data quote – values of one eye at two months or values of both eyes at one month ; <table border="1"> <thead> <tr> <th>time / months</th><th colspan="2">change in visual acuity / au</th></tr> <tr> <th></th><th>eye not treated</th><th>eye treated</th></tr> </thead> <tbody> <tr><td>0</td><td>0.00</td><td>0.00</td></tr> <tr><td>1</td><td>-0.12</td><td>-0.20</td></tr> <tr><td>2</td><td>+0.02</td><td>-0.40</td></tr> <tr><td>3</td><td>0.00</td><td>-0.50</td></tr> <tr><td>4</td><td>-0.13</td><td>-0.52</td></tr> <tr><td>5</td><td>-0.22 or -0.23</td><td>-0.52</td></tr> <tr><td>6</td><td>-0.15</td><td>-0.54</td></tr> <tr><td>7</td><td>-0.12</td><td>-0.50</td></tr> <tr><td>9</td><td>-0.15</td><td>-0.33</td></tr> <tr><td>10</td><td>-0.08</td><td>-0.52</td></tr> <tr><td>12</td><td>-0.12</td><td>-0.54</td></tr> </tbody> </table>	time / months	change in visual acuity / au			eye not treated	eye treated	0	0.00	0.00	1	-0.12	-0.20	2	+0.02	-0.40	3	0.00	-0.50	4	-0.13	-0.52	5	-0.22 or -0.23	-0.52	6	-0.15	-0.54	7	-0.12	-0.50	9	-0.15	-0.33	10	-0.08	-0.52	12	-0.12	-0.54	3
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3(c)(ii)	any three from: 1 mutated DNA is, deleted / removed / cut out or DNA replaced with normal DNA ; Ignore exon X is deleted as this is mRNA not DNA Reject gene deleted / cut out 2 specific (locations) / between exon 26 and exon 27, (in the genome) ; 3 functional protein / CEP290, now made OR correct functioning of photoreceptors ; 4 acts on person's own DNA ; 5 AVP ; e.g. STOP codon no longer present	3

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